New York House Price Predictions

Abstract

Multiple regression models enable the prediction of numerical outcomes using two or more predictor variables. In this study, we analyse a 2006 housing dataset from Saratoga County, New York, to develop a multiple regression model for predicting house prices based on key property features. Our analysis identifies eight significant features that influence housing prices: living area, land value, number of bathrooms, waterfront status, age, new construct status, heat type and lot size.

Introduction

Our study aimed to build a predictive model of house prices (USD) and to identify which home features are most strongly associated with price. The linear models we developed, mainly multiple regression, can act as a statistical tool for price prediction and provide substantial insights into the drivers of house value. These findings may then inform decision-making for buyers, sellers and housing analysts, answering questions like “Is this property undervalued?”.

Data Set

The dataset was accessed from the Data and Story Library. It was a random sample of 1734 houses taken from the full Saratoga Housing Data dataset, containing data on houses in Saratoga County, New York, USA in 2006. This dataset contains 17 variables, listed in *Figure 1* of the appendix. We assume samples are independent due to the way the data was collected.

Analysis

To assess the suitability of a multiple linear regression model, we examined the scatterplots from our exploratory data analysis. The approximately linear relationships observed among most numerical variables, including Price, supported the linearity assumption required for regression analysis. However, we found that applying a log transformation to Price enhanced linearity across predictors. Consequently, all subsequent modelling was performed using log(Price) as the response variable.

To identify the most appropriate set of predictors, we applied forward and backward stepwise selection as well as an exhaustive search, using the Akaike Information Criterion (AIC) as the model selection criterion. The forward and backward procedures produced identical models, as seen in *Figure 2*, and the exhaustive search, using the AIC criterion, confirmed the same nine predictors as optimal. However, based on the p-values in *Figure 2* that test for the significance of the slope predictor, the variable Pct.College was excluded as it was not statistically significant.

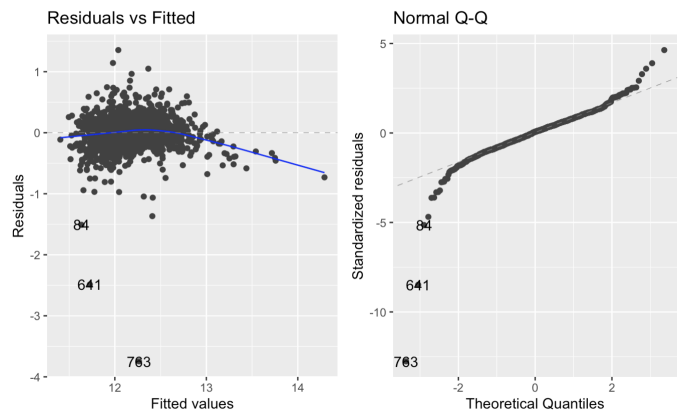

Figure 3: Residual and QQ plot from our final model.

To ensure that the linearity, homoskedasticity, and normality assumptions for multiple regression were met, we compared the residuals vs. fitted and Q-Q plots shown in *Figure 3*. The residuals plot shows points symmetrically scattered around zero with an approximately constant spread, suggesting that both the linearity and homoskedasticity assumptions are reasonably satisfied. Although slight deviations appear in the tails of the Q–Q plot, the majority of points fall along the reference line. Given our large sample size (n = 1734), the Central Limit Theorem ensures that the normality assumption is adequately met.

To address potential heteroskedasticity, we tested several transformations of the response and predictor variables, including logarithmic and power transformations; however, these adjustments did not yield any notable improvement. Outliers identified in *Figure 3* were examined and found to be plausible data

points representing legitimate housing prices, so they remained in the analysis.

The correlation matrix in *Figure 4* indicates that multicollinearity among predictors is minimal, with all pairwise correlations below 0.7. For each categorical variable, (k–1) dummy variables were included to prevent multicollinearity arising from the inclusion of redundant categories.

Finally, the robustness of our selected model was validated using the variable inclusion plot (*Figure 7*) and the model stability plot (*Figure 8*). Figure 7 demonstrates that the predictors in our final model have a consistently high probability of selection and are well separated from the redundant-variable boundary, indicating their importance in predicting house prices. Figure 8 further confirms that lot size, one of our chosen predictors, enhances model stability, and our 8-variable model achieves both high accuracy and stability across resamples.

Results

Our multiple linear regression model included eight predictor variables (*Figure 6*). As indicated by the R^2 values, our model accounts for 57.2% variance in log(Price) based on 1299 observations. The model's out-of-sample performance was assessed using 10-fold cross-validation on test data. As demonstrated in *Figure 5*, our model has a mean R^2 of 0.583 and an RMSE of 0.291, indicating the model performs well on both in-sample and out-of-sample data. *Figure 6* shows that highly significant predictors ($p \leq 0.001$) were Living Area, Land Value, Bathrooms, Waterfront, Age, New Construct and Lot Size. For HeatType, only 'Hot Air' was statistically significant. 'Hot Water' ($p = 0.099$) and 'None' ($p = 0.114$) were not.

Conclusion

Our study allows us to conclude that house prices in Saratoga County, New York (in 2006) are driven by many factors. Most significantly, Waterfront properties are associated with the largest positive effect on price (Estimate = 0.45), and Living Area, Land Value, Bathrooms, and Lot Size all show a significant positive association with log(Price).

	Log Price	
Predictors	*Estimates*	*p*
(Intercept)	11.24	**<0.001**
Living Area	0.00	**<0.001**
Land Value	0.00	**<0.001**
Bathrooms	0.11	**<0.001**
Waterfront	0.45	**<0.001**
Age	-0.00	**<0.001**
New Construct	-0.19	**<0.001**
Heat Type [Hot Air]	0.08	**<0.001**
Heat Type [Hot Water]	0.05	0.099
Heat Type [None]	-0.22	0.114
Lot Size	0.04	**<0.001**
Observations	1299	
R^2 / R^2 adjusted	0.572 / 0.569	

Figure 6. Final multiple regression model with estimates and p-value for each predictor.

Discussion

Limitations

One limitation is the assumption that relationships between price and the predictor variables are linear. Furthermore, stepwise selection procedures for model building do not explore all model combinations, whereas an exhaustive search does. Additionally, the dataset contained data from a single location in 2006, reducing its generalizability to other time frames or locations.

Future Research

From our individual exploratory data analysis and *Figure 4*, we observe that the relationships between housing variables may be better modelled with non-linear relationships. Models such as a random forest or k-nearest neighbours could be explored. A random forest model incorporates categorical variables more naturally, whilst k-nearest neighbours can easily adapt to new data, characteristics that are both desirable for a model aimed at predicting house prices. Furthermore, due to the significant impact that economic conditions have on housing prices, other variables such as interest rates or inflation could be incorporated as predictor variables.

Appendix

Numerical Variables

- Price (Target)
- Lot.Size
- Age
- Land.Value
- Living.Area
- Pct.College

Categorical Variables

- Waterfront (Binary)
- New.Construct (Binary)
- Central.Air (Binary)
- Fuel.Type (Nominal)
- Heat.Type (Nominal)
- Sewer.Type (Nominal)
- Bedrooms (Discrete)
- Fireplaces (Discrete)
- Bathrooms (Discrete)
- Rooms (Discrete)

Figure 1: A list of all variables from the original dataset.

	Forward model		**Backward model**	
Predictors	*Estimates*	*p*	*Estimates*	*p*
(Intercept)	11.17	**<0.001**	11.17	**<0.001**
Living Area	0.00	**<0.001**	0.00	**<0.001**
Land Value	0.00	**<0.001**	0.00	**<0.001**
Bathrooms	0.11	**<0.001**	0.11	**<0.001**
Waterfront	0.47	**<0.001**	0.47	**<0.001**
Age	-0.00	**<0.001**	-0.00	**<0.001**
New Construct	-0.17	**<0.001**	-0.17	**<0.001**
Heat Type [Hot Air]	0.08	**<0.001**	0.08	**<0.001**
Heat Type [Hot Water]	0.05	0.107	0.05	0.107
Heat Type [None]	-0.22	0.102	-0.22	0.102
Lot Size	0.04	**<0.001**	0.04	**<0.001**
Pct College	0.00	0.085	0.00	0.085
Observations	1299		1299	
R^2 / R^2 adjusted	0.573 / 0.569		0.573 / 0.569	
AIC	517.754		517.754	

Figure 2: Forward and Backward stepwise search models.

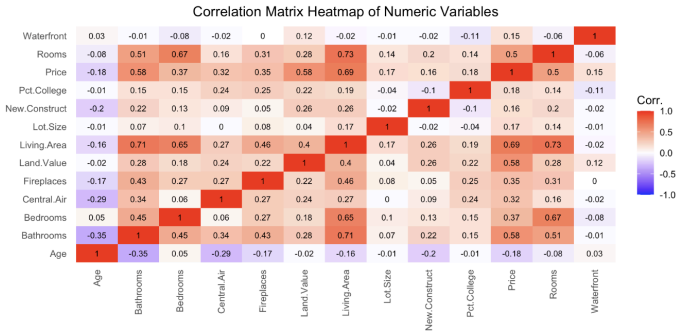

Figure 4: Correlation matrix heatmap to assess for multicollinearity.

```
Linear Regression

1299 samples
   8 predictor

No pre-processing
Resampling: Cross-Validated (10 fold)
Summary of sample sizes: 1168, 1169, 1169, 1170, 1169, 1169, ...
Resampling results:

  RMSE      Rsquared   MAE
  0.291441  0.5830611  0.2094234

Tuning parameter 'intercept' was held constant at a value of TRUE
```

Figure 5: k-fold cross-validation results for k = 10.

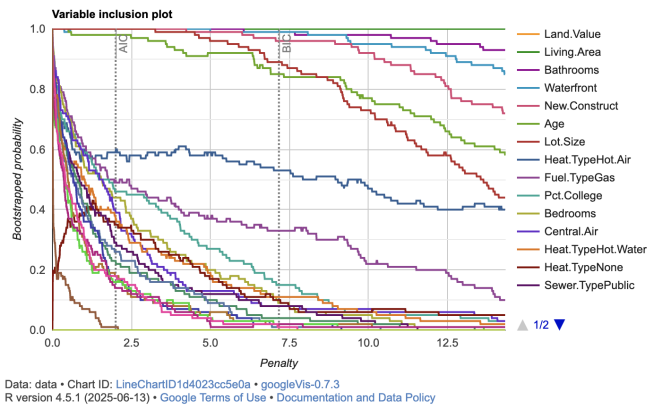

Figure 7: variable inclusion plot

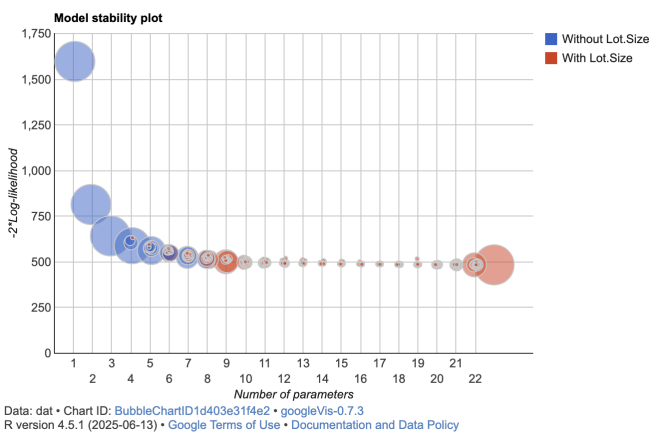

Figure 8: model stability plot